PRACTICAL NO.6(B)

Aim:- Apply the concept of expected value to evaluate different decision alternatives and select the optimal one.

Code:-

```
high_demand_probability=0.60
low_demand_probability=0.40
#probabilities and payoffs
high_demand_profit=80000
low_demand_profit=20000
do_not_launch_profit=30000
ev_launch=(
 high_demand_probability*high_demand_profit
 +low_demand_probability*low_demand_profit)
ev_not_launch=do_not_launch_profit

#Display result
print("DECISION MAKING UNDER UNCERTAINITY")
print("-----------------------------------")
print("Expected value- Launch Product:", ev_launch)
print("Expected Value- Do Not Launch:", ev_not_launch)
#select the best decision
if ev_launch> ev_not_launch:
 print("Best Decision: Launch Product")
elif ev_not_launch> ev_launch:
 print("Best Decision: Do not Launch")
else:
 print("Both Decisions have the same expected value.")
```

Output:-

```
DECISION MAKING UNDER UNCERTAINITY
----------------------------------
Expected value- Launch Product: 56000.0
Expected Value- Do Not Launch: 30000
Best Decision: Launch Product
```

Concluison:-

The Decision Tree model successfully demonstrated how business decisions can be made under uncertainty by considering different possible outcomes and their associated probabilities. By calculating the **Expected Value (EV)** for each decision alternative, the outcomes were compared objectively. The alternative with the **highest expected value** was identified as the optimal decision. Thus, the practical showed that decision trees and expected value analysis help businesses reduce uncertainty, evaluate risks, and make more informed and profitable decisions.